

Interference in Text Categorisation Experiments

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Abstract. This article examines manual textual categorisation by human coders with the hypothesis that the law of total probability may be violated for difficult categories. An empirical evaluation was conducted to compare a one step categorisation task with a two step categorisation task using crowdsourcing. It was found that the law of total probability was violated. Both a quantum and classical probabilistic interpretations for this violation are presented. Further studies are required to resolve whether quantum models are more appropriate for this task.

1 Introduction

Automatic categorisation of documents is gathering interest from researchers in text processing and machine learning and the development of novel approaches to solve this problem has led to a number of annotated collections. Such a paradigm generally assumes that although there can be disagreement on a category label for a given document, each document labelled in a series is processed independently of others, and each potential category for a document is considered in isolation. Therefore the development and the evaluation of automatic categorisation algorithms is deeply dependent on this assumption. However it has been found that some categories may be problematic to such algorithms: that is they are difficult to label properly. In this paper, we conjecture that such a situation may arise from a default of the assumption of independence between documents and categories, and propose to study this from a decision making perspective.

The study of the relationship between perception, categorisation and decision making is an active research area since the beginning of 2000. Townsend et al. [1] introduced a new paradigm to study these type of interactions in order to test Markov models. The framework used is the following: (i) a stimulus is presented to a subject, (ii) the subject is then asked to categorise it, (iii) finally the subject is asked to make a decision. The research question investigated is whether the probability of a decision depends on the characteristics of the initial stimulus, or is the probability of the decision affected by the categorisation step? In other words, can a conscious cognitive process that does not impact the reality of the stimulus interfere with a decision about the stimulus? The hypothesis was that

subjects would follow a Markov property, that is, the final decision is based on the categorisation step and not on the original stimulus.

The hypothesis was tested in an experiment, in which subjects were shown pictures of faces and asked to categorise them into one of two possible groups and thereafter asked to decide on a course of action. The two groups were defined by the following visual features: face shape (wide, narrow); lip thickness (thick, thin). The main procedure of the experiment was divided into two parts. In Part I, subjects were asked to:

1. categorise a picture of a face (usually indicated as the input signal S) in either group G (good guys) or group B (bad guys);
2. decide on an action A (act aggressively) or F (act friendly), knowing that one group (B in this case) is more hostile than the other.

In Part II, subjects only had to make the decision response, not both.¹ subjects were given information about the correct proportions of face shapes for each category (about 60 %) and they were given incentives for each correct answer.

The corresponding Markov model underpinning this experiment is given by:

$$P(A|S) = P(A|G)P(G|S) + P(A|B)P(B|S) \quad (1)$$

$$P(F|S) = P(F|G)P(G|S) + P(F|B)P(B|S) \quad (2)$$

It was found that the law of total probability was not satisfied: the probability of taking a decision without categorising the face (left hand side of Eqs. 1 and 2) was significantly different from the decision taken after the categorisation step (right hand side of Eqs. 1 and 2). This is the consequence of the fact that most individuals are moderately influenced by their categorisation when they have to make a decision.

The quantum interaction research community has presented evidence that such empirical violations of the law of total probability can be modelled with quantum-like models using an interference term. The interference term modifies the law of total probability and is derived from incompatibility between sub-spaces representing events. A quantum-like inference model for this experiment was proposed [2,3] and such models have been widely proposed in quantum cognition [4–8].

The aim of this article is to investigate whether the law of total probability is being violated in text categorisation, and if it is, to investigate whether it can be modelled using a quantum-like interference model.

2 Interference in Text Categorisation

Text categorisation is the task of deciding whether a piece of text belongs to one or more categories, given a set of specified categories [9]. This task has

¹ In the experiment, the reverse order, decide-then-categorise, and categorise only was investigated as well. We do not analyse the reverse order in this paper.

important applications in the real world, for example: news stories are typically organised by subject categories or geographical areas; academic papers are often classified by domains and research areas; email messages are classified into the two categories of spam and non-spam. Since manual text categorisation is a time consuming process, machine learning approaches are used to automate this task. For supervised learning algorithms and for the evaluation of the automatic classifiers, some ground truth is required. In general, the ground truth is a collection of textual documents sampled from the same population of documents that will be automatically classified. This collection is carefully annotated by experts of the topics (or the categories) covered by the documents.

The goal of a supervised machine learning algorithm is to learn a classifier based on the manually annotated documents. More formally, given a finite set of pre-defined categories $C = \{c_1, \dots, c_i, \dots, c_n\}$ and a finite set D of documents, the learning of a categorizer consists of the definition of a function for each category c_i of this kind: $\Phi_i : D \rightarrow [0, 1]$. Therefore, a categorizer returns a degree of membership of a document d for a category c_i .² However, when the decision to assign or not d to c_i has to be taken, a threshold t may be needed such that if $\Phi_i > t$, d is assigned to category c_i , otherwise it is not assigned to c_i . The function Φ_i takes on different meanings according to the learning method used: in the Naïve Bayes approach, it is defined in terms of the conditional probability that a document d belongs to the category c_i , $P(c_i|d)$. This approach of learning a classifier for each class is called binary categorisation. Given a set of categories C , there are $|C|$ independent binary categorisation problems where each document of D must be assigned either to category c_i or its complement $\bar{c}_i = C - c_i$.

In machine learning literature, it has been consistently found that it is harder to learn a classifier for some categories. Our hypothesis for this phenomenon is that when human annotators decide that a document belongs to category c_1 , they are necessarily undecided about whether it belongs to category c_2 . This indecision is the hallmark of incompatible subspaces, which have been shown not to adhere to the law of total probability [6, 8].

The process of categorising documents is very close to the categorisation and decision making problem described in the Sect. 1. More specifically, the act of categorising a document is *both* a categorisation step, i.e., a document is categorised into category c , and a decision step, i.e., placing the document in the ‘box’ which contains the documents of category c . Therefore, the process of categorisation can be modelled with a Markov chain, as shown in Fig. 1, in the following way:

- given a set of categories C and a document d to categorise;
- step 1, categorise d in either category c_1 or $\bar{c}_1$;
- step i , categorise d in either c_i or $\bar{c}_i$, knowing that the document was previously categorised under either c_{i-1} or $\bar{c}_{i-1}$.

The hypothesis here is similar to the original experiment of [1]: a human expert would follow a Markov property, that is the decision at the i -th step is

² The image of Φ_i can be different, for example $\Phi_i : D \rightarrow \mathbb{R}$.

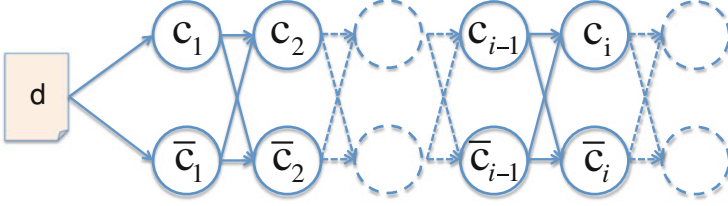


Fig. 1. The process of categorisation described by a Markov chain.

based on the categorisation information used at step $i - 1$ and not on the features of the original document.

2.1 Categorisation Under Two Categories

In order to test the hypothesis using the face detection experimental framework, we contain the categorisation task to a task of deciding upon only two categories, $C = \{c_1, c_2\}$. In this new experiment, subjects will be asked to classify documents into either c_1 or not, and into either c_2 or not. The experiment is divided into two parts. In Part I, subjects will be asked to: (i) first, categorise a document in either category c_1 or $\bar{c}_1$; (ii) then, categorise the document under either c_2 or $\bar{c}_2$. In Part II, subjects will be asked to only decide whether the document is relevant for either c_2 or $\bar{c}_2$.

Given a document d , by the law of total probability, the probability of categorising d under c_2 or $\bar{c}_2$ is:

$$P(c_2|d) = P(c_2|c_1)P(c_1|d) + P(c_2|\bar{c}_1)P(\bar{c}_1|d) \quad (3)$$

$$P(\bar{c}_2|d) = P(\bar{c}_2|c_1)P(c_1|d) + P(\bar{c}_2|\bar{c}_1)P(\bar{c}_1|d) \quad (4)$$

In Part I of the experiment, the probabilities of the right hand side of the last two equations are estimated; while in Part II, the probabilities of the left hand side, $P(c_2|d)$ and $P(\bar{c}_2|d)$, are computed.

In order to be able to use the same formalism for analysing the outcomes as the face recognition experiment, the design of the text categorisation experiment must be as similar to it as practical. Therefore we need to carefully address the following issues: (1) the choice of features that describe the category and (2) the proportion of documents for each category.

In [1], and subsequently in [2], the features of the type of the faces of the two groups (good guy or bad guy) were clearly identified as a two dimensional space: roundness of the face (round or narrow), thickness of the lips (thick or thin). The face had to have neither any expression nor any hair style (all the faces were actually bald). Half of the faces were round with thin lips, half were narrow with thick lips. The size of these two features was slightly altered to have a range of different faces. The values of the size of the face and lips thickness was such that each group could be clearly identified. In this experiment, these two features

were the only hint that could drive the judgement of the subject. Additionally, subjects are informed that although these are the general describing features of each category of face, they are neither necessary nor sufficient conditions. This is described further in the description of proportions of documents in categories.

For a text categorisation experiment, the situation is more complex. Text documents are high-dimensional objects since they are represented by words. Categories, in turn, can be described by many words or concepts that are related to the documents of the category. For this reason, the categorisation experiment cannot be limited to two dimensions.

The second issue concerns the proportion of documents between categories. The original experiment was designed to be completely symmetric, that is the altered faces were assigned to the two groups in the following way: round faces with thin lips had a 60 % chance of being assigned to the bad guy group; bad guys had a 70 % chance of being hostile. The other category had symmetric chances.

In the case of text categorisation, categories are very often unbalanced; therefore, building an experiment with the same number of positive and negative examples for each category would be quite unrealistic. In general, a document that presents the features of a category (we address the problem of the features in Sects. 2.2 and 2.4) belongs to that category; nevertheless, there are some documents that seem to cover a topic of a category but belong to the opposite category. We decided to manually select a reasonable sample of documents, by reasonable we mean comparable to the original experiment (34 pictures), from a standard benchmark such that: (i) a document may belong to both categories, to either c_1 or c_2 , or to none; (ii) a document that belongs to a category (according to the classification given by the standard benchmark) may or may not contain the keywords used to summarise the category; (iii) a document that belongs neither to c_1 nor to c_2 (according to the classification given by the standard benchmark) may contain some of the keywords used to summarise the categories of interest.

2.2 Materials

The Reuters-21578 has been the most important benchmark for text categorisation since the 1990s. It consists of 21,578 news articles that appeared on the Reuters newswire in 1987. The articles were manually categorised by experts. By examining the description of the creation of the original dataset [10], it is possible to build some analogies with the Markov text categorisation process. For example, news stories had been hand-categorised for any of 72 categories available, but it was required to assign at most 6 of the 72 categories for each document. A story could be assigned one or more of these codes, or no code at all if none of the chosen six was appropriate.³ The important step that clarifies the link between the Markov process and is that experts had to select from a

³ The restriction to six codes was imposed to keep the effort required to build the dataset within certain budgetary limits.

set of categories for each document. Since, as humans, we need to process each decision in sequence, it is very likely that the experts had in mind a specific order of the 72 categories. Therefore, they were actually performing a sort of 72-step experiment, with the extra constraint of a maximum of 6 categories per news.

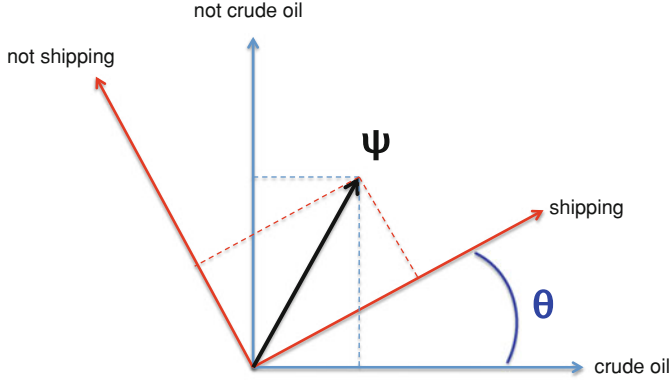


Fig. 2. Incompatibility between the two subspaces “shipping” and “crude oil”. ψ denotes the cognitive state of the subject with respect to document d , the projections on the axis are the probabilities that the subject decides for one category or the other. θ measures the interference between the two subspaces.

Table 1. A breakdown of the 82 documents selected for the experiment.

Only ship	Only crude	Crude and ship	Not crude and not ship
20	14	29	19

There is also another important issue that reflects the problem of choosing the right features to describe categories. Quoting [10]: “Closer examination of the results, however, shows that the kind of errors made are quite different. Human errors stem mainly from inconsistent application of categories, especially the categories with the vaguest definitions, and from failing to specify all the categories when several should have been assigned to a story.”

The process of manually categorising the Reuters collection presents many of the issues that we are addressing in this paper. For the two categories experiment, we select two Reuters categories: (1) a category about “shipping”, and (2) a category about “crude oil”. These two categories are good candidates for this pilot experiment because they have distinctive keywords that characterise each topic. However, there are documents that are about shipping crude oil and there are documents that mention ship and crude without being categorised in either category. For this reason, we hypothesise there may be incompatibility between the subspaces corresponding to “shipping” and the other corresponding to “crude oil” as depicted in Fig. 2. See Sect. 3 for further analyses about incompatibility.

After a careful manual selection, a total of 82 documents were selected. Table 1 shows the number of documents for each combination.

Table 2. Number of observation and number of subjects per group.

Number of HITS approved		500	1000	5000	Masters
Observations	One step task	796	790	734	681
	Two step task	802	813	808	829
# of Workers	One step task	30	42	48	45
	Two step task	32	62	38	42

2.3 Subjects

For this experiment, we used the Mechanical Turk⁴ crowdsourcing platform for recruiting subjects. Collecting relevance judgements via crowdsourcing using MTurk has a demonstrated track record and has been shown to be of equal quality as that of lab-based experts [11, 12]. For the purposes of this experiments, we decided that unit of work to be performed by a worker (a HIT (Human Intelligence Task)) would be based on the study of a single document (and either one or 2 categorisation questions). We followed a similar crowdsourcing methodology advocated by [13].

A variable that was manipulated was the “Number of HITS approved”, that is the experience of the workers. In Mechanical Turk, there are seven levels: from 0 (anyone can participate) to “masters” (best workers). We ran the experiments with four levels of expertise to see whether there is any difference in the results between these groups. The levels are (the figure indicates the number of approved HITS): 500, 1000, 5000, and masters. For each experiment, we have a total number of observations equal to 82 documents times 10 observations per document. We report in Table 2 the exact number of observations after data cleansing (HIT not approved, inconsistent answers, workers who categorised the same documents more than once in the same experiment).

HITS were published in batches on Mechanical Turk from December 2012 until January 2013. For each level of expertise, the one step and the two step categorisation tasks were published at the same time. We started from the group of workers with at least 500 HITS approved. When all the HITS were completed for both tasks, we published the HITS for the workers of the next level. For workers at level masters it was a bit different, since we found masters to be more difficult to involve. For this reason, we had to publish the HITS more than once and add one cent to the base pay rate as an incentive (this is also why we have less observations for the one step categorisation).

⁴ <https://www.mturk.com/>

2.4 Procedure

One Step Categorisation. During a one step categorisation HIT, a document was presented with the following layout:

- Description: The goal of this HIT is to decide which categories short news articles belong to.
- Instructions: (i) Read the news article carefully. (ii) The article consists of a title of one line (capital letters) and a body.
- Content: the actual content of one of the 82 REUTERS document is shown in a box. We showed only the tags TITLE and BODY of the original document.
- Decision: Decide whether the news is about "shipping" given to the following clues:
 - Such news often mention the size of the cargo being shipped, e.g., "tonnes" or related words such as "port";
 - Sometimes these news mention what is being shipped, e.g., "oil" or "crude";
 - Bear in mind, not all documents that contain such words necessarily belong to the category "shipping".

At this point, the worker is asked whether the document is about "shipping" (Y/N).

Two Step Categorisation. During a two step categorisation HIT, a document is presented with the same initial layout as the one step categorisation. The difference is in the decision part:

- First Decision: Decide whether a news article belongs to the category "crude oil" or not, e.g., such news often mentions related words like "barrel" and/or "oil", but not all the documents that contain these words belong to the category "crude oil".
- Question: Is this document about crude oil? (Y/N)
- Second decision: Decide whether the news is about "shipping" or not, e.g., such news often mentions related words such as "tonnes" and/or "port" frequently but not all the documents that contain these words are about shipping. Bear in mind that documents about "crude oil" have the tendency to be about "shipping" too, but this is not always the case.
- Question: Is this document about shipping? (Y/N)?

2.5 Results

A breakdown of the results, in the same manner as [2], is given in Table 3. The first part of the table, indicated with 'Two step', presents the results of the estimate of the probabilities for each transition of the Markov model in the case of the two step classification task. The second part, 'One step', is the estimate of the probabilities of the one step classification task. The last column, 'Reuters', is the actual probability of category c_2 if we consider the Reuters tags. For each

Table 3. Experimental results Part I. Category c_1 corresponds to “crude oil”, category c_2 to “shipping”. We indicated in bold the values that are closer to the original Reuters classification.

Level	Two step						One step	Reuters
	Type	$P(c_1 d)$	$P(c_2 c_1)$	$P(\bar{c}_1 d)$	$P(c_2 \bar{c}_1)$	$P(c_2)$	$P(c_2)$	
500	c_1	0.7442	0.3250	0.2558	0.3364	0.3279	0.4466	0.3256
	$\bar{c}_1$	0.2231	0.4217	0.7769	0.4671	0.4570	0.5173	0.5323
1000	c_1	0.7459	0.3219	0.2541	0.3303	0.3240	0.4552	0.3240
	$\bar{c}_1$	0.2214	0.2588	0.7786	0.4950	0.4427	0.5055	0.5208
5000	c_1	0.7687	0.3647	0.2313	0.3434	0.3598	0.5431	0.3271
	$\bar{c}_1$	0.2263	0.4651	0.7737	0.5170	0.5053	0.6147	0.5263
Masters	c_1	0.7964	0.3551	0.2036	0.3667	0.3575	0.2514	0.3258
	$\bar{c}_1$	0.2558	0.4747	0.7442	0.5382	0.5220	0.3823	0.5323

level of expertise, we split the results according to what category the document given to the user belongs to, that is documents that belong to either category c_1 or not (this corresponds to showing pictures of good guys or bad guys).

Let us begin the analysis by observing the results of the first part of the two step categorisation task. In theory, the probability $P(c_1|d)$ of the first step should be very close to 1 (or close to 0 for documents that are not in c_1) because we know that we are giving the workers only the documents that belong to c_1 (or $\bar{c}_1$). Nevertheless, this probability never reaches 0.80 in either case and it is almost symmetric. This behaviour may indicate that there are some documents that are difficult to classify and the worker randomly assigns the documents to either c_1 or not.

The second part of the two step categorisation task allows us: (i) to estimate the marginal probability of c_2 , and (ii) to compare it with the one computed in the one step categorisation task, and (iii) to compare it with the original Reuters manual categorisation. There is an interesting pattern that emerges from this part of the analysis which we highlighted in bold in Table 3. Unexperienced workers (with less than 5,000 HITS) tend to categorise the correct proportion of c_1 documents when they perform the two step categorisation task, while they over-estimate it in the case of the one step task. Conversely, they underestimate the proportions of $\bar{c}_i$ documents in the two step task, while they almost correlate perfectly with the Reuters’ manual coders in the one step task. Experienced workers (more than 5,000 HITS) are very close to the correct proportions for both c_i and $\bar{c}_i$ when they perform a two-step categorisation task, while there is an over/under-estimation for the one step case.

These results suggest that a two step categorisation task may model the original Reuters labelling process faithfully. In fact, 6 out of 8 times the proportion of documents of category c_2 of the two-step experiment was very close to the Reuters classification, and it was almost perfectly matched in the case of more experienced users.

3 Discussion

The experiment shows that during manual text categorisation humans are affected by the order of the categories. Therefore, the assumption that each potential category for a document is considered in isolation is violated. Since we are interested in finding a mathematical framework for text categorisation that captures this effect, we begin this discussion by assuming that incompatibility exists in the document classification and the law of total probability is therefore being violated. By incompatibility, we mean that the joint probability between the categories in the two-step decision task cannot be formed.

The Naïve Bayes classifier is used as the focal point of discussion, which has the following form:

$$\Pr(c_1|d) \propto \Pr(c_1)\Pr(d|c_1) \quad (5)$$

The law of total probability with respect to category c_2 can be brought in as follows:

$$\Pr(c_1|d) \propto [\Pr(c_1, c_2) + \Pr(c_1, \bar{c}_2)]\Pr(d|c_1) \quad (6)$$

which we argue is violated, i.e.,

$$\Pr(c_1|d) \propto [\Pr(c_1, c_2) + \Pr(c_1, \bar{c}_2) + Intf]\Pr(d|c_1) \quad (7)$$

where *Intf* is an interference term. We can express this as follows in terms of the geometry of incompatible subspaces:

$$\Pr(c_1) = \|\mathbf{P}_1\psi\|^2 \quad (8)$$

$$= \|(\mathbf{P}_1 \cdot \mathbf{I})\psi\|^2 \quad (9)$$

$$= \|(\mathbf{P}_1 \cdot (\mathbf{P}_2 + \mathbf{P}_2^\perp))\psi\|^2 \quad (10)$$

$$= \|\mathbf{P}_1\mathbf{P}_2\psi\|^2 + \|\mathbf{P}_1\mathbf{P}_2^\perp\psi\|^2 + Intf \quad (11)$$

where ψ denotes the cognitive state of the subject with respect to document d , $\|\mathbf{P}_1\psi\|^2$ is the probability that the subject decides for category c_1 . Similarly, $\mathbf{P}_2$ and its dual $\mathbf{P}_2^\perp$ model that the subject decides for category c_2 (or not). In this way, the interference term can be integrated with the standard Naïve Bayes classification model.

The advantage of the quantum-like model is that it offers a clean mathematical framework which both motivates and defines the additional parameter *Intf*. From a machine learning point of view, the variable *Intf* is a parameter of the model that can be estimated from past observations. If the misjudgements are consistent, this parameter may be used as a correction factor of the bias introduced by humans, hence we may have a new automatic classifier that predicts/corrects future decisions. In our case, we can fit the data almost perfectly in two of the experiments which are: (i) workers with at least 5,000 HITS and documents that do not belong to c_1 , (ii) master workers and documents that

Table 4. Experimental results Part II. Fitting the data to the quantum model.

Level	Two step		$P(c_2 c_1)$	$P(\bar{c}_1 d)$	$P(c_2 \bar{c}_1)$	θ
	Type	$P(c_1 d)$				
5000	$\bar{c}_1$	0.23	0.48	0.77	0.52	1.3
Masters	$\bar{c}_1$	0.25	0.47	0.75	0.53	1.9

do not belong to c_1 . Compare the fitted values of Table 4 to the estimates in Table 3.

A drawback of this model is that we introduce one constraint: the matrix of transition probabilities from one state to another (that is from one decision to another) must be double stochastic. However, in many experimental data, this condition is not met [14]. Even in the original experiment of faces [2], not all the results could be explained by a quantum model. In addition, some may question whether the law of total probability is actually being violated. In fact, a classical probabilistic model describes in the best possible way the evidence we have of an experiment. Consider the experiment with two questions. We have a sample space that is $\Omega_2 = \{(c_1, c_2), (\bar{c}_1, c_2), (c_1, \bar{c}_2), (\bar{c}_1, \bar{c}_2)\}$. In this space, the only way to compute the marginal probability of c_2 is, by following the classical axioms, $Pr_{\Omega_2}(c_2) = Pr_{\Omega_2}(c_1, c_2) + Pr_{\Omega_2}(\bar{c}_1, c_2)$.⁵ Therefore, when we observe this experiment, we cannot claim a violation of the law of total probability.

The misunderstanding about this violation arises when we observe a second experiment, that is the one question experiment with a sample space is $\Omega_1 = \{c_2, \bar{c}_2\}$. At this point, the fact that the two probabilities are different, $Pr_{\Omega_1}(c_2) \neq Pr_{\Omega_2}(c_2)$, is nothing more than an ‘unexpected’ fact. It is like tossing one coin and computing the probability of heads and tails, then we toss the same coin together with another coin. We ‘expect’ to obtain the same probability of heads and tails because there is no reason to think otherwise. If we do not, it is not because the LTP is violated but because there is something we could not predict.⁶

4 Summary and Conclusions

The results of the pilot experiment presented in this paper accords with other similar studies in quantum cognition namely (1) Markov models are not sufficient to model multiple categorisation decisions, and (2) in some instances quantum-like interference models may be a superior model. The immediate consequences of such findings are that there is reason to believe that evaluation collection in text categorisation are biased and therefore provide a substandard account of human manual categorisation. What is more worrying there is that many

⁵ The underscript Ω_2 reminds us in which space the probabilities are computed.

⁶ For example, think about two loaded dice that contain a small magnet inside. Tossed separately, they work as fair dice; tossed together, the magnetic field influences the outcome.

categorisation algorithms base their processing (training for machine learning algorithms) on these biased versions of the truth. We propose that larger scale experiments should shed more light on the specifics of this bias and provide some insights into how interference terms should be uncovered.

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